

Exploring the Role of Digital Traditional Music Platforms in Cross-Cultural Music Pedagogy

Qian Zhang

 <http://orcid.org/0009-0006-6660-874X>

School of Art and Design, Huaqing College, Xi'an University of Architecture and Technology, China

Received: September 23rd, 2025 | **Accepted:** March 25th, 2026

ABSTRACT

This study proposes and evaluates a culturally adaptive digital platform for traditional music education in cross-cultural settings. Using a design-based research approach, the platform integrates multimodal resources, multilingual interfaces, and real-time feedback to enhance learner engagement, cultural understanding, and emotional resonance. Empirical results from diverse learners show improved cognitive mastery and strong emotional response, although participation in discussions requires further enhancement. The findings highlight the potential of technology-enhanced solutions in preserving intangible cultural heritage and promoting intercultural empathy in global educational environments. Future work will focus on artificial intelligence personalization and broader cultural scalability.

KEYWORDS

Traditional Music, Digital Teaching Platform, Intercultural Education, Multimodal Resources, Cultural Integration

INTRODUCTION

Traditional music, as a vital component of intangible cultural heritage, encapsulates not only artistic forms but also the historical consciousness, spiritual beliefs, and social customs of distinct cultural communities. As both a cultural archive and a dynamic mode of expression, traditional music reflects the collective identity and lived experiences of its people. It plays a central role in cultural continuity, moral education, and communal rituals, rendering it a significant pedagogical and anthropological resource in education (Benford et al., 2021; Joseph et al., 2020). However, in recent decades, the transmission of traditional music has encountered unprecedented disruption due to globalization, urbanization, and the dominance of Western pedagogical systems in formal education.

In this evolving context, digital technologies have emerged as both a challenge and an opportunity. On one hand, the digitization of education has often led to the marginalization of culturally rooted forms of knowledge. On the other hand, it provides powerful tools for the preservation, dissemination, and revitalization of traditional music through multimedia representation, virtual collaboration, and interactive learning environments (Hamutoglu et al., 2020). These technologies allow for broader access and greater flexibility, enabling learners across linguistic and geographic boundaries to engage with musical traditions that might otherwise be inaccessible. Digital platforms, if thoughtfully designed, offer the potential to mediate between authenticity and accessibility, allowing traditional

DOI: 10.4018/IJICTE.407363

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music to thrive within a global learning ecosystem. This observation aligns with Cao (2025), who shows that short-form social media platforms such as TikTok can meaningfully increase young audiences' exposure to—and emotional attachment toward—intangible musical heritage. These findings reinforce the argument that digitally mediated cultural content can accelerate cross-cultural reach and youth engagement.

Cross-cultural education has become an increasingly prominent context in which traditional music is being taught. Within such environments, learners come from diverse sociolinguistic and cultural backgrounds, often possessing different cognitive styles, aesthetic preferences, and prior knowledge bases (Kang, 2022). This diversity introduces both pedagogical opportunities and challenges. On the one hand, it enables intercultural dialogue and the promotion of global citizenship. On the other hand, it risks cultural dilution, superficial engagement, or misinterpretation of culturally embedded meanings. A critical tension thus arises between the need to preserve cultural authenticity and the imperative to make content comprehensible and engaging for diverse learners. This tension is particularly acute in the case of non-Western musical traditions, which often operate within radically different tonal systems, rhythmic frameworks, and performance conventions compared to standardized Western models (Allotey-Pappoe, 2025; Zhan, 2025).

Traditional methods of music instruction—such as oral transmission, apprenticeship, and face-to-face mentorship—have historically been effective in preserving the cultural integrity of musical forms. These approaches emphasize holistic learning, including the acquisition of context-specific knowledge, emotional intuition, and community values. However, they are often constrained by limitations of scale, location, and instructor availability. As educational institutions and cultural organizations increasingly turn to digital solutions, the question becomes: how can we replicate or reinterpret the immersive, culturally grounded nature of traditional instruction in online or hybrid settings?

Several recent studies have explored the integration of digital tools into music education, particularly in the context of formal schooling and university-level programs. These efforts have ranged from the development of multimedia content and virtual instruments to the use of artificial intelligence (AI) in music analysis and personalized instruction (Yu et al., 2023). Nevertheless, significant gaps remain. Many platforms prioritize technical skill acquisition—such as pitch accuracy, rhythm, and instrumental dexterity—while overlooking the cultural and emotional dimensions of music learning. Others adopt a Eurocentric design ethos that fails to adequately represent the structure and symbolism of non-Western traditions. Moreover, few platforms address the intercultural cognitive mismatches that arise when learners interpret music outside their native cultural frameworks.

Given these shortcomings, there is a pressing need for digital learning environments that are explicitly designed to support cross-cultural engagement with traditional music. Such platforms must go beyond functional digital delivery to offer meaningful cultural immersion, interpretive scaffolding, and real-time feedback mechanisms. They must also be responsive to diverse learning needs, offering adaptive content, semantic supports, and user-friendly interfaces that accommodate a range of digital proficiencies.

This study addresses these challenges by proposing and evaluating a digital platform for traditional music education tailored to cross-cultural contexts. The platform employs a modular, multilingual architecture that supports multimodal content delivery, including audiovisual demonstrations, annotated music scores, interactive exercises, and formative assessments. It also integrates culturally sensitive design elements, such as symbolic prompts, localized narratives, and emotional computing features, to foster deeper engagement and understanding. In this study, “emotional computing features” specifically means:

- Chinese/English bilingual sentiment polarity analysis (SnowNLP + TextBlob) of learners' reflection logs;

- according to the proportion of negative emotions > 0.4 , the content adaptation of “encouraging video + reducing difficulty” is automatically triggered;
- the teacher's background dashboard displays the class emotional heat map in real time, which is used to adjust the teaching rhythm.

The research adopts a design-based research (DBR) methodology, which allows for iterative development and testing of the platform in real-world educational settings. Empirical data were collected from a diverse cohort of learners through system logs, user interaction data, questionnaires, and interviews. The study evaluates the platform's effectiveness in four key dimensions: cognitive mastery of musical concepts, cultural understanding, emotional engagement, and interactive participation. Special attention is given to how learners from different cultural backgrounds respond to culturally embedded content and how digital tools mediate their interpretive processes. These findings are consistent with Petkova (2025), who similarly observed that culturally contextualized digital prompts can increase learners' interpretive depth when engaging with traditional arts. However, while Petkova's study focused primarily on cognitive scaffolding, the present work extends this direction by incorporating emotional computing features, demonstrating that affective resonance further amplifies cross-cultural comprehension.

The objectives of this study are threefold: to design a culturally adaptive digital platform that effectively supports traditional music instruction in cross-cultural environments; to empirically assess the platform's impact on learner outcomes across multiple dimensions; and to offer a theoretical and practical framework for future research and development in digital music pedagogy and intercultural education.

Despite the increasing interest in digitally supported traditional music education, existing research presents several gaps. First, most digital platforms emphasize content delivery but lack culturally adaptive mechanisms capable of responding to learners' emotional and cognitive states. Second, few empirical studies have examined how cross-cultural learners interact with traditional music materials in digital environments, limiting understanding of cultural variability in learning processes. Third, prior work rarely integrates emotional engagement, interaction behaviors, and cultural comprehension into a unified analytical framework. Addressing these gaps, this study develops an emotionally informed, culturally adaptive digital platform and provides empirical cross-cultural evidence that advances current technological and pedagogical research in digital music education.

By integrating technological innovation with cultural authenticity, this study aims to contribute to the fields of educational technology, digital humanities, and arts education. It offers a model for how digital platforms can be leveraged not only to disseminate knowledge but also to preserve and transmit the intangible cultural assets that define human diversity. In doing so, it addresses broader questions about the role of technology in cultural sustainability and the potential for education to serve as a bridge across cultural divides.

RELATED WORK

The integration of digital technologies into music education has been a prominent research focus over the past two decades, particularly in relation to cultural preservation, learner engagement, and pedagogical innovation. Within this body of literature, the role of traditional music in cross-cultural learning environments has received increasing scholarly attention. However, existing studies often remain fragmented across disciplinary boundaries—spanning education, musicology, cognitive science, and human-computer interaction—resulting in a lack of coherent frameworks for the design and evaluation of culturally responsive digital platforms. This section reviews key strands of relevant literature to situate the present study and identify critical gaps.

Traditional Music and Cultural Transmission

Traditional music has long been recognized as a vehicle for transmitting cultural values, historical memory, and communal identity. Scholars emphasize that such music is embedded in specific social practices and belief systems, often functioning as a living form of heritage (Xiao, 2025). As Nikkanen and Westerlund (2017) note, the pedagogical significance of traditional music extends beyond musical technique to include rituals, narratives, and symbolic meaning. Recent research continues to extend this trajectory. For instance, Simatupang et al. (2025) demonstrate that augmented reality/virtual reality–based gamified environments significantly enhance learners’ willingness to engage with traditional music practices, providing empirical evidence that immersive technologies can function not only as instructional tools but also as mechanisms for cultural preservation.

In nonformal learning contexts, traditional music is often passed down through oral transmission, mentorship, and participatory performance. These methods emphasize experiential learning, emotional resonance, and communal participation. However, their effectiveness depends heavily on proximity to cultural communities and access to knowledgeable bearers of tradition—conditions that are increasingly rare in urbanized and globalized settings.

To address the erosion of such traditions, scholars and practitioners have begun to explore how digital platforms can serve as tools for both preservation and revitalization. This includes digitizing archives, producing educational documentaries, and developing interactive applications. However, while these efforts often succeed in conserving content, they frequently fail to replicate the immersive and relational aspects of traditional learning.

Digital Music Education and Multimodal Learning

The application of educational technology in music learning environments has expanded dramatically in recent years. Researchers such as Liu et al. (2022) and Duraisamy (2025) advocate for multimodal learning environments that integrate audio, visual, textual, and kinesthetic modalities to improve knowledge retention and learner engagement. These multimodal resources include instructional videos, annotated sheet music, animations, and interactive practice tools.

From a cognitive perspective, multimodal design aligns with dual-coding theory and cognitive load theory, which posit that learning is more effective when information is processed through multiple channels (Li, 2024). In music education, this approach facilitates both conceptual understanding and sensory immersion, particularly in contexts involving abstract symbolic systems such as notation or rhythm.

Nevertheless, most existing digital music platforms are geared toward Western classical or popular music traditions. As Konovalova et al. (2025) point out, such platforms often assume tonal systems, harmonic structures, and pedagogical methods rooted in Western paradigms, which may not align with the features of non-Western or indigenous music traditions. The result is a lack of cultural fidelity and pedagogical appropriateness for learners seeking to engage with alternative musical forms.

Cross-Cultural Music Pedagogy and Cultural Sensitivity

Cross-cultural music education aims to promote intercultural understanding, empathy, and global citizenship through exposure to diverse musical traditions. Li et al. (2024) introduce the concept of cultural resonance, which suggests that learners must first develop cultural empathy before acquiring technical musical skills. This approach prioritizes contextual understanding, emotional engagement, and symbolic interpretation.

In digital environments, cultural sensitivity can be supported through various means: interface design that reflects cultural aesthetics, semantic scaffolding to explain symbolic references, and multilingual content delivery (Sihombing et al., 2021). For example, Palmer et al. (2022) emphasizes that culturally responsive pedagogy in music education should include not only content diversity but also culturally congruent instructional strategies.

Yet, despite increasing awareness of the need for cultural sensitivity, many digital platforms lack systematic strategies for addressing intercultural cognitive mismatches. Learners may misinterpret cultural signs, fail to recognize stylistic nuances, or experience emotional disconnection. These challenges underscore the need for systems that support real-time feedback, contextual interpretation, and emotional resonance.

Platform Adaptability and Learner-Centered Design

Recent innovations in educational technology have shifted toward learner-centered and adaptive systems. Meta-learning frameworks, as discussed by Prest (2020), emphasize real-time feedback loops, behavioral analytics, and content customization. These features are particularly important in cross-cultural contexts, where learners' prior knowledge, motivations, and learning styles vary significantly.

Adaptive platforms can enhance inclusivity by providing differentiated pathways, semantic aids, and personalized learning sequences. For example, Sterner (2022) notes that East Asian learners often benefit from contextual explanations and cultural comparisons, which help reduce ambiguity and enhance relevance. Similarly, De Bruin (2021) advocates for emotion-aware systems that monitor learner affect and respond with appropriate interventions.

Despite these advancements, few digital music platforms have fully incorporated adaptive learning mechanisms tailored to cross-cultural users. Most remain static in their content delivery and lack sufficient scaffolding for interpreting cultural elements. Moreover, data-driven personalization is often underutilized in music pedagogy compared to disciplines such as language learning, science, technology, engineering, and math.

Empirical Evaluation and Learner Outcomes

Several studies have attempted to evaluate the impact of digital platforms on music learning outcomes. Metrics typically include skill acquisition, learner satisfaction, and engagement rates. For example, Wu and Qin (2025) measured emotional engagement in a digital music education system using affective computing techniques and found a strong correlation between emotional immersion and cultural understanding.

Other researchers, such as Yu et al. (2023), have employed gamified systems to assess the effectiveness of visual and interactive tools in enhancing music literacy. However, there remains a paucity of research that integrates cognitive, emotional, and cultural metrics in a unified evaluation framework. This is particularly true in cross-cultural settings, where outcomes must reflect not only technical proficiency but also intercultural sensitivity and emotional resonance.

Identified Research Gaps

Based on the literature reviewed, the following research gaps are identified:

- Lack of culturally adaptive platforms: most digital music learning systems do not incorporate cultural cues, symbolic annotations, or localized pedagogical models necessary for authentic engagement with traditional music.
- Insufficient empirical studies on cross-cultural learners: there are limited data on how learners from different cultural backgrounds interpret and respond to traditional music content in digital environments.
- Underutilization of adaptive learning technologies: existing platforms rarely apply AI-based learner profiling, real-time analytics, or affective computing to personalize cultural learning experiences.
- Limited emotional and symbolic scaffolding: emotional engagement is often neglected, despite its critical role in music appreciation and cultural immersion.

- Disconnection between cultural theory and platform design: few studies effectively integrate theoretical models of intercultural competence, heritage education, and music semiotics into platform architecture.

MATERIALS AND METHODS

This study adopts a DBR approach to explore the development and application of a digital platform for traditional music education in cross-cultural settings. The methodology consists of three main components: platform architecture design, multimodal resource integration, and user interaction modeling. To provide visual support and further clarification, this section also includes figures and tables relevant to each technical module.

Participants

A total of 156 learners participated in this study. The sample represented six major cultural and linguistic backgrounds, including the East Asian, Southeast Asian, Middle Eastern, European, African, and Latin American regions. Participants ranged in age from 18 to 32 years old, with most enrolled in undergraduate or master-level programs. Recruitment was conducted through open enrollment via partner universities offering cross-cultural music education courses.

All learners were assigned to the same instructional conditions using the digital platform, consistent with the DBR framework. No random-group assignment was performed, as the primary goal of this research phase was exploratory system evaluation rather than controlled comparative testing. These characteristics provide foundational diversity for examining culturally driven differences in interaction and engagement.

Platform Architecture Design

The platform interface follows the layout of three columns and one floating layer: the left longitudinal navigation column (multilingual switching, progress badge), the central main column (music score + cultural annotation pop-up window), the right feedback column (real-time pitch curve + percentage correct), and the bottom solid input area (reflection log). All elements are embedded in a single page and can be operated without scrolling the screen.

The platform is structured using a multilayered modular architecture. It includes three primary layers: the resource management layer, the instructional interaction layer, and the user support layer:

- The resource management layer organizes and manages audio, video, musical notation, and contextual materials. Resources are indexed and classified for efficient access and pedagogical coherence.
- The instructional interaction layer supports synchronous and asynchronous teaching activities, including task assignments, real-time feedback, and communication tools. This layer enables dynamic interaction between teachers and students.
- The user support layer includes learner profiles, activity logs, and personalization tools, allowing for customized content delivery and progress tracking.

The system also incorporates multilingual support and cultural context prompts to improve accessibility for users from different backgrounds. The interface blends traditional cultural aesthetics with modern usability principles, enhancing user engagement.

To illustrate the system logic, Figure 1 presents the architectural flowchart of the platform, and Table 1 summarizes the core functions of each module.

Figure 1. Traditional Music Digital Teaching Platform System Structure Diagram

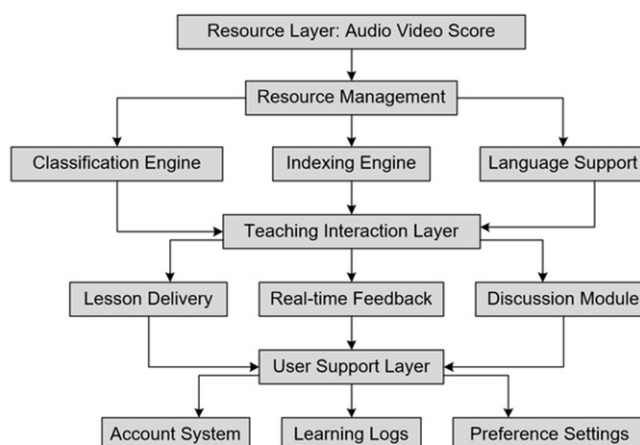


Table 1. Main Platform Function Modules and Descriptions

Functional Module	Description
Audio/Video Management	Unified access to multisource media content
Multilingual Support	Interfaces and cultural guidance in multiple languages
Real-Time Feedback System	Supports interactive and formative assessment during learning
Learning Path Tracking	Monitors learner behavior to recommend adaptive content

Multimodal Resource Integration

To preserve the richness of traditional music, the platform incorporates multimodal resources: audio clips, instructional videos, music scores, and interactive exercises. These resources are aligned with four sensory modalities: auditory, visual, textual, and experiential:

- Audio and video form the core of content delivery, facilitating auditory recognition and cultural immersion.
- Music scores are used to reinforce theoretical understanding and performance accuracy.
- Interactive browser-based tools simulate musical practice and provide instant feedback, encouraging learner participation.
- Resources are organized into a unified database and mapped to specific course units. Cultural annotations accompany each resource to contextualize learning and minimize cultural misinterpretation.

Figure 2 presents a schematic view of the multimodal teaching resource framework, and Table 2 matches resource types to instructional scenarios.

Figure 2. Schematic Diagram of Multimodal Teaching Resource Integration

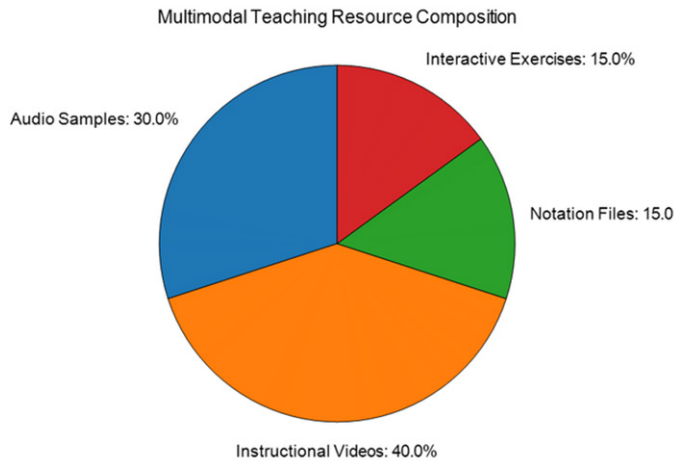


Table 2. Comparison of Teaching Resource Types and Application Scenarios

Teaching Resource Type	Functional Focus	Application Scenario
Audio Samples	Aural identification, performance style analysis	Timbre and rhythm training
Instructional Videos	Demonstrative content and cultural narratives	Scene reenactment, guided learning
Sheet Music Files	Theoretical reinforcement, notation practice	Structure analysis, melody reproduction
Interactive Exercises	Feedback-driven learning and skill practice	Improvisation, assessment, performance prep

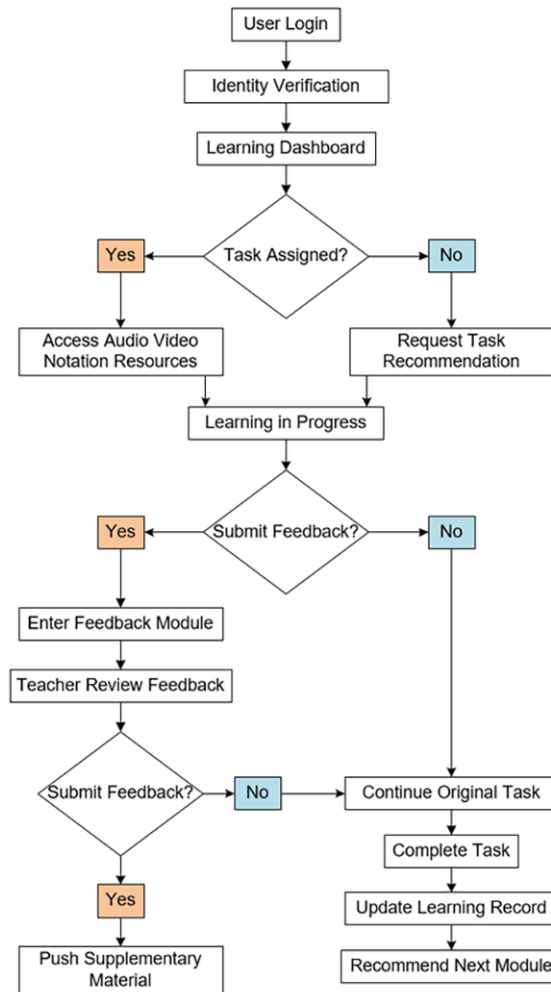
Interaction and Feedback Design

The platform’s interaction design follows a closed-loop model of input–interaction–feedback–adjustment. User engagement begins with task recommendations based on learning history. Learners can submit responses via comments, quizzes, or performance uploads. Teachers respond with formative feedback or additional resources.

A key innovation is the integration of cultural sensitivity monitoring. The system uses cues such as unfamiliar terminology, symbolic prompts, and real-time assistance to help learners identify and address cognitive blind spots. This supports a more inclusive and responsive learning environment.

Figure 3 demonstrates the full cycle of user interaction, including cultural prompt modules and teacher feedback mechanisms.

Figure 3. User Interaction and Feedback Closed-Loop Flowchart



By combining these technical and pedagogical strategies, the platform aims to bridge cultural gaps and promote meaningful engagement with traditional music in diverse educational contexts. This methodological framework lays the foundation for the practical results and evaluations discussed in subsequent sections.

Beyond describing interaction frequency, Figure 3 provides evidence of how learners' engagement trajectories align with the platform's instructional design. The repeated peaks in comment and annotation activity suggest that culturally contextualized tasks successfully triggered reflective learning moments. This pattern supports the study's objective of fostering sustained cultural engagement rather than surface-level task completion.

To support cross-cultural interpretation, the platform also logged learners' cultural background categories (e.g., regional grouping and linguistic heritage). These data were incorporated into the descriptive analyses to observe potential differences in participation patterns across cultural subgroups, although no formal statistical subgroup comparison was conducted at this stage.

Definition of Evaluation Metrics

To improve clarity, the operational definitions of the evaluation indicators are provided below:

- **Comment activity index:** this index reflects the level of learner interaction within the platform. It is calculated based on the frequency of comments posted, the distribution of comments across course units, and the ratio of active commenters to total learners. A score of 69.5 indicates moderate participation, with approximately two to four comments per active learner per module.
- **Emotional response score:** emotional engagement was measured through a 5-point Likert-scale questionnaire administered after each learning module. Items assessed learners' levels of interest, immersion, and emotional resonance with the content. Scores were standardized to a 100-point scale, where higher values indicate stronger emotional involvement.
- **Keyword matching degree:** this metric represents the alignment between learners' open responses and the predefined set of cultural and conceptual themes within the course, measured using semantic similarity algorithms and manual verification.
- **Video completion rate:** indicates the percentage of assigned audiovisual materials completed by learners, weighted by duration and viewed segments.

These clarifications enhance the transparency of the evaluation process and strengthen the interpretability of the results.

Qualitative Data Analysis

In addition to quantitative indicators, qualitative data were collected through semi-structured interviews and open-ended questionnaire responses. A thematic analysis approach was applied. Two researchers independently performed open coding, grouped similar codes into categories, and resolved discrepancies through discussion to ensure coding consistency. Themes were derived based on frequency, conceptual relevance, and cross-cultural patterns reflected in the data. Representative excerpts from participants' responses were anonymized and incorporated to illustrate key findings and provide interpretive depth.

Statistical Analysis

Given that the current study did not include a structured pretest/posttest design or multiple experimental iterations, inferential statistics comparing learning gains were not applicable. Therefore, the analysis focused on exploring relationships among post-use indicators, including cognitive mastery, cultural understanding, interaction participation, and emotional engagement.

Pearson correlation analysis was applied to examine potential associations between learners' engagement behaviors and learning outcomes (e.g., whether higher participation was linked to stronger cultural understanding). All analyses were conducted to provide preliminary quantitative insight, acknowledging that causal inference is not possible within the current design.

Technical Performance and Usability Evaluation

The platform's technical performance was monitored throughout the deployment. System logs showed an uptime rate of 99.2% during the study period, with no major interruptions affecting instructional progress. Minor interface latency issues reported during the first DBR cycle were addressed through front-end resource optimization.

Usability feedback was also collected through open-ended questionnaire items. Learners and instructors highlighted ease of navigation, clarity of module layout, and responsiveness of the feedback panel as positive aspects. Suggestions included increasing the visibility of cultural annotations and simplifying access to the interactive exercise panel. These inputs informed iterative refinements made between DBR cycles, ensuring that the system interface evolved in response to user needs.

Contribution of the Present Study

This study seeks to address these gaps by developing and empirically evaluating a digital platform designed specifically for traditional music education in multicultural contexts. It offers a holistic framework that integrates:

- culturally annotated multimodal resources;
- adaptive learning modules informed by user behavior and emotional feedback;
- multilingual and symbolic interface design;
- cross-cultural evaluation metrics capturing cognitive, emotional, and participatory outcomes.

By drawing on interdisciplinary theories from digital pedagogy, musicology, and intercultural communication, the platform moves beyond static content delivery to offer a dynamic, responsive, and culturally grounded learning environment.

RESULTS

This section presents the application outcomes of the proposed digital platform in a cross-cultural teaching environment. Data were collected from platform logs, student interaction records, feedback responses, and evaluation reports. The analysis focuses on three core areas: curriculum structure implementation, cross-cultural interaction dynamics, and multidimensional learning assessment.

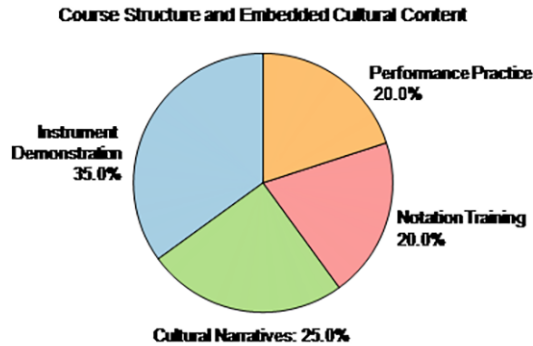
Curriculum Design and Cultural Integration

The platform's curriculum framework comprises four core modules, instrument demonstration, cultural narrative, score training, and performance practice, sequentially arranged to move learners from basic recognition to cultural comprehension and expressive performance:

- Instrument demonstration: provides audiovisual presentations of traditional instruments to build foundational familiarity.
- Cultural narrative: offers stories, symbols, and contextualized subtitles to connect musical forms to heritage.
- Score training: combines regional notation with standard staff notation to support dual literacy.
- Performance practice: simulated and recorded performances allow students to apply learning outcomes.

Figure 4 illustrates the course architecture and its embedded cultural components.

Figure 4. Curriculum Structure and Cultural Embeddedness Model Diagram



The system enables content customization based on user profiles. Learners receive appropriate annotations and content levels, ensuring personalized cultural immersion. Cultural prompts are embedded throughout, helping students develop sensitivity and avoid misinterpretation.

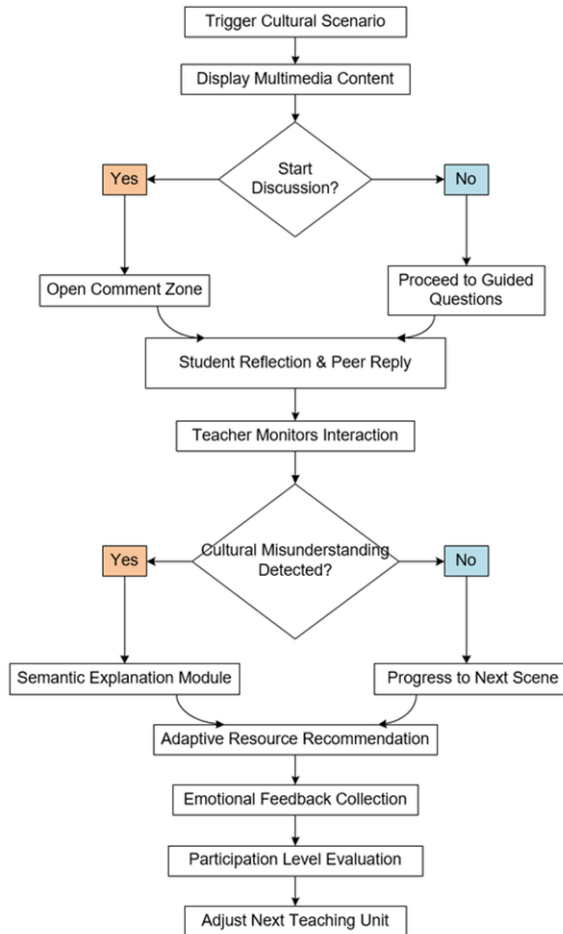
Cross-Cultural Interaction Mechanism

Interactive features are designed to stimulate cultural curiosity and encourage knowledge exchange. The interaction path includes contextual triggers, exploratory learning, feedback loops, and guided reflection.

Students interact via comment threads, situational question and answer sections, and virtual performance spaces. Teachers use data analysis tools to assess comprehension depth and adjust teaching resources accordingly. The platform supports semantic breakdown of cultural concepts, visual comparisons, and multilingual annotations.

Figure 5 visualizes the cross-cultural interaction process.

Figure 5. Flowchart of Cross-Cultural Interaction Path



These interactive features enhance engagement and reduce cultural misreading. Emotional and behavioral indicators are collected to evaluate participation quality and adjust strategies. The distribution shown in Figure 5 highlights that emotional resonance is not merely a by-product of exposure but a key driver of cultural interpretation. Higher emotional response scores correlate with deeper cultural comprehension, reinforcing the finding that affective engagement plays a mediating role in cross-cultural learning. This directly supports the study's aim of integrating emotional computing features to enhance cultural understanding.

Learning Effectiveness Evaluation

The evaluation framework comprises four dimensions: cognitive mastery, cultural understanding, interactive participation, and emotional engagement. Metrics are collected through system analytics and user feedback.

Table 3 shows typical evaluation results based on platform data generation. In terms of cognitive mastery, the average score of students' video completion rate is 87.3 points and the compliance rate is as high as 91.2%, indicating that the basic knowledge teaching effect is good. In the cultural understanding dimension, the keyword matching score is 78.6 points, and the compliance rate is 82.4%. Although it is high, it is slightly weaker than the cognitive dimension, indicating that there is still room for

improvement in students’ in-depth grasp of cultural semantics. In terms of interactive participation, the comment activity index score is 69.5, which shows that some students are not active enough in expressing their views or participating in the discussion. The emotional response score of the emotional engagement dimension is 84.1 and the compliance rate is 88.5%, which reflects that students’ emotional resonance with music content is generally high, especially in the teaching of cultural contexts other than their mother tongue. This result is of particular reference significance.

Table 3. Evaluation Results From the Cross-Cultural Teaching Platform (Sample Size = 156)

Evaluation Dimension	Indicator Name	Average Score (out of 100)	Compliance Rate (%)
Cognitive Mastery	Video Completion Rate	87.3	91.20
Cultural Understanding	Keyword Matching Degree	78.6	82.40
Interactive Participation	Comment Activity Index	69.5	75.00
Emotional Engagement	Emotional Response Score	84.1	88.50

To further examine relationships among the measured indicators, a correlation analysis was conducted across the four dimensions. Results showed a moderate positive correlation between interactive participation and cultural understanding ($r \approx 0.42$), indicating that learners who engaged more actively in discussion and feedback tended to achieve higher cultural comprehension scores. Emotional engagement was also positively associated with cultural understanding ($r \approx 0.37$). These exploratory correlations suggest that participation intensity and emotional resonance may contribute to deeper cultural learning, although causal interpretations cannot be made within the current study design.

The results indicate strong performance in content acquisition and cultural empathy. While participation in discussions was slightly lower, emotional and cultural dimensions showed high levels of resonance. These findings support the platform's effectiveness in fostering holistic learning experiences.

Additionally, automated reports generated by the system help instructors identify performance trends and adjust strategies, reinforcing a dynamic and responsive teaching environment.

Qualitative Findings

Three major themes emerged from the thematic analysis:

- Educators’ digital literacy influenced learner experience. Participants noted variation in response timeliness and feedback quality depending on instructors’ technological proficiency. “Some teachers responded quickly and used the platform features well, but others seemed unfamiliar with the tools, which affected how smoothly the activities ran” (Participant No. 12).
- Cultural familiarity shaped emotional engagement. Learners reported stronger resonance when content was related to their own cultural background. “I felt more connected when the music was from my region, but the annotations helped me understand unfamiliar traditions too” (Participant No. 41).
- Real-time feedback increased motivation. Learners emphasized that immediate system responses and teacher comments encouraged more active participation.

These qualitative insights complement the quantitative results by highlighting how human factors such as teacher readiness and cultural proximity influence platform effectiveness.

DISCUSSION

The results of this study strongly support the hypothesis that a culturally adaptive digital platform can significantly improve the teaching and learning of traditional music within cross-cultural educational environments (Tang, 2022). These findings offer valuable contributions to both practical applications and theoretical advancements in digital education and music pedagogy, emphasizing the critical importance of integrating cultural responsiveness into technology-enhanced learning systems.

The cultural heterogeneity of the learner cohort further strengthens the interpretation of the results. Because the sample included learners from six distinct cultural regions who demonstrated comparable engagement trends, the findings provide preliminary evidence that the platform's culturally adaptive features function effectively across diverse user groups.

The inclusion of qualitative findings further enriches the interpretation of the results. Thematic analysis of interview and open-ended responses revealed that factors such as educators' digital literacy and learners' cultural familiarity shaped their engagement behaviors. These qualitative themes provide important human-centered insight that complements the quantitative metrics and helps contextualize platform performance in real teaching scenarios.

The iterative improvements observed across DBR cycles, especially those based on user feedback regarding interface clarity and annotation accessibility, demonstrate that the platform design was responsive to practical classroom needs and contributed to its overall usability and stability.

From the standpoint of cultural representation and preservation, the platform effectively maintained the authenticity of local musical aesthetics and their rich historical backgrounds (Xie, 2024). This was accomplished through the deliberate incorporation of multimedia demonstrations (including audiovisual content), comprehensive cultural annotations, and a dual-notation approach designed to accommodate diverse musical literacies (Lee & Lin, 2023). Such localized content serves to reduce cultural misinterpretations and facilitates the development of foundational cultural schemas among learners who are initially unfamiliar with the musical traditions presented. By embedding cultural elements not only within the instructional content but also throughout the platform's user interface and curriculum design, the system transcends the role of a mere teaching tool and becomes an active agent in the preservation and dissemination of intangible cultural heritage through digital means.

In terms of learner engagement, the platform's interactive features demonstrated considerable efficacy. Real-time comment systems, personalized feedback, and exploratory learning tasks fostered active learner participation and reflection. However, the data revealed uneven levels of engagement in certain areas, particularly within comment-based discussion forums. This variability indicates the need for additional motivation-enhancing strategies such as peer collaboration activities, gamification components (including badges, leaderboards, and achievement recognition), and social acknowledgment mechanisms to promote sustained and consistent learner involvement. Beyond motivation-based explanations, the uneven comment participation can also be interpreted through social presence theory, which suggests that learners' willingness to contribute in online settings depends on perceived instructor presence, peer visibility, and interaction norms. In low-presence environments, learners tend to hesitate before initiating or sustaining discussion, even when cognitively engaged. Strengthening social cues—such as guided prompts or structured community tasks—may therefore increase discussion participation more effectively than motivational interventions alone.

Emotional engagement metrics were notably high, especially in modules emphasizing immersive storytelling and culturally significant symbolism. This finding aligns with existing literature (e.g., Zhang & Chen, 2025), which highlights emotional immersion as a critical catalyst for deepening cultural understanding and enhancing educational outcomes in intercultural learning contexts. Emotional resonance, therefore, emerges as a pivotal gateway through which learners can connect more profoundly with the content, enabling richer and more meaningful learning experiences. These findings are consistent with Pekrun's academic emotions framework, which argues that positive

activating emotions such as interest and curiosity enhance cognitive processing and promote deeper cultural comprehension. In the context of traditional music learning, emotional resonance supports interpretive sensitivity, enabling learners to form empathetic connections with culturally embedded musical narratives. This connection also aligns with theories of cultural empathy development, where affective engagement acts as a gateway to intercultural understanding.

From a technical perspective, the platform's modular architecture and robust multilingual support significantly enhance its adaptability across diverse learner populations. Nonetheless, challenges remain in effectively addressing the needs of learners at the extremes of the musical expertise spectrum—from beginners to advanced musicians. Future iterations of the platform should prioritize the integration of AI-driven learner profiling techniques capable of delivering more granular, personalized content recommendations tailored to individual proficiency levels and learning preferences. Differences in learner responses across cultural groups can also be understood through intercultural sensitivity theory, which proposes that learners' readiness to interpret unfamiliar cultural phenomena varies according to their developmental stage of cultural awareness. Learners with lower intercultural sensitivity may rely more heavily on explicit scaffolding, while those with higher sensitivity demonstrate stronger interpretive autonomy. The platform's culturally annotated modules and multilingual explanations appear to function as scaffolds that support learners progressing through these stages.

Another critical consideration identified in this study is the role of instructors within the digital learning ecosystem. The effectiveness of the platform was significantly influenced by educators' digital literacy and their capacity to adapt pedagogical methods to a dynamic, culturally diverse digital environment (Petkova, 2025). Therefore, ongoing professional development initiatives focusing on enhancing teachers' digital competencies and culturally responsive teaching strategies are essential to facilitate widespread and effective adoption of such platforms (Yao & Li, 2023).

While this research concentrated on a specific selection of traditional music forms, future work should explore the scalability and adaptability of the platform across a broader range of cultural and regional contexts (Tang & Zeng, 2024). Expanding the musical repertoire and incorporating language support for underrepresented and minority music traditions—including those of indigenous peoples—would substantially increase the platform's inclusiveness, cultural richness, and global relevance.

A key limitation of this study is the absence of a formal control group or baseline comparison, which restricts the ability to precisely quantify the platform's improvement relative to traditional classroom instruction. Although the user-profiling questionnaire offered preliminary insights into learners' prior knowledge, it did not operate as a structured pretest suitable for statistical comparison.

To enhance methodological rigor, future research will incorporate a two-group experimental design, including a control group receiving conventional face-to-face instruction and an experimental group using the proposed platform. A standardized pretest/posttest evaluation will also be designed to compare cognitive mastery, cultural understanding, and emotional engagement across instructional conditions. Such comparative data will allow more reliable measurement of the platform's effectiveness and strengthen the generalizability of the conclusions.

Despite the positive outcomes, several limitations should be acknowledged to contextualize the findings. First, the cultural scope of the current platform is relatively narrow, as the implemented modules cover only a small number of musical traditions. This restricts the generalizability of the results to broader intercultural contexts. Expanding the platform to include a more diverse range of musical heritages would allow for a more comprehensive examination of cross-cultural learning outcomes.

Second, the study did not employ a controlled experimental design. Without a formal control group or baseline comparison, the extent to which learning gains can be attributed solely to the platform—rather than to external factors or prior learner knowledge—cannot be fully quantified. Future research should incorporate pretest/posttest measures, controlled groups, or comparative designs with traditional instruction to strengthen causal claims.

In summary, the findings underscore the significant potential of the platform as an innovative cross-cultural educational technology. By addressing core challenges in digital music pedagogy—

namely cultural authenticity, learner engagement, and adaptive, culturally informed instruction—the platform offers a comprehensive, interactive solution that aligns pedagogical goals with evolving technological affordances. Continued development and research efforts should focus on enhancing this synergy to maximize the platform’s educational impact and applicability in increasingly diverse and digitally mediated learning environments.

CONCLUSION

This study proposed and empirically evaluated a digital teaching platform specifically designed to preserve and disseminate traditional music within cross-cultural educational contexts. By integrating multimodal resources, culturally embedded curriculum frameworks, and adaptive interaction mechanisms, the platform successfully fostered cognitive understanding, cultural appreciation, emotional engagement, and active learner participation. Its modular, multilingual, and feedback-driven architecture proved effective in supporting learners from diverse backgrounds in acquiring both technical musical skills and deeper cultural literacy.

The findings contribute a robust framework to the field of digital music education, demonstrating the critical importance of embedding cultural narratives, symbolic representations, and emotion-driven content into digital learning environments. These design elements collectively mitigate cultural barriers, facilitating more inclusive and meaningful learning experiences while improving overall educational outcomes. Additionally, the platform’s ability to evoke emotional resonance with unfamiliar musical traditions highlights the potential of technology-mediated approaches to enhance intercultural empathy and engagement.

Despite these promising results, several limitations warrant further attention. The current platform would benefit from the incorporation of advanced AI-powered personalization to better address the diverse proficiency levels of learners. Variability in learner engagement signals the need for motivation-enhancing features, such as gamification and structured peer collaboration models, to sustain consistent participation. Furthermore, instructor digital readiness emerged as a pivotal factor influencing the platform’s pedagogical effectiveness, underscoring the necessity for comprehensive teacher training and professional development programs.

Future research should aim to broaden the empirical application of the platform across diverse cultural and linguistic contexts, especially focusing on underrepresented and indigenous musical traditions, to verify and enhance its global scalability. Investigations into affective computing and emotional analytics could also deepen the platform’s pedagogical sophistication, enabling real-time adaptation to learner emotional states. Moreover, the development of targeted teacher-training modules will be crucial to support educators in effectively integrating this technology into culturally responsive teaching practices.

In conclusion, this study confirms that technology-mediated traditional music education holds substantial promise for fostering cultural preservation, pedagogical innovation, and cross-cultural exchange in increasingly multicultural learning environments. By harmonizing cultural depth with adaptive instructional design, the proposed platform offers a sustainable and scalable pathway to revitalize and share musical heritage in the digital age, aligning with the evolving demands of global education. Future research will incorporate more extensive qualitative sampling and mixed-methods triangulation to deepen the understanding of how cultural background and instructional practices influence learner outcomes.

DATA AVAILABILITY

The figures used to support the findings of this study are included in the article.

COMPETING INTERESTS

The author of this publication declares there are no competing interests.

FUNDING STATEMENT

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. Funding for this research was covered by the author of the article.

ACKNOWLEDGMENT

The author sincerely thanks those who have contributed to this research.

CORRESPONDING AUTHOR

Correspondence should be addressed to Qian Zhang: 13909182209@163.com.

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Qian Zhang, Female, was born in Xi'an, Shanxi, China, in February 1986. She received her Master's degree in Music Education from Xi'an University of Architecture and Technology. Currently, she works as a Lecturer at the College of Art and Design, Huaqing College, Xi'an University of Architecture and Technology. She has participated in one provincial-level teaching reform project and one ministerial/provincial-level longitudinal research project. Her research interests include Solfège (Ear Training) and Piano Performance.